



Apple Nets 40 million dollars

Apple's Recycling Program mints \$40 Million in Gold from unwanted or broken apple devices.
<http://digit.in/Apple40m>

Microsoft adds Offline Translation

The latest version of Microsoft Translation supports offline translation with a simple download of offline language packs.
<http://digit.in/Moffline>

From the labs



VPU will keep your smartphones as slim as ever

This could help in two things -- the first, in enhancing camera quality without making your phone thicker. One of the main reasons why smartphones can't reach DSLR quality are their space constraints. You can not fit large enough lenses or sensors into them. VPUs, potentially, can solve this. Secondly, it could also improve low light photography, a major area of focus for smartphone OEMs. While a lot of advancements have been made by companies like Apple and Samsung, low light remains the bane of smartphone cameras, and VPUs may help here as well.

Brains and brawns

Perhaps the most interesting and potentially scary implementation of VPUs though is in machine learning. A neural network of machine learning algorithm replicates the human brain and a VPU can

act as the eyes for that brain. On January 27, 2016, Movidius announced that it is working with Google to accelerate the adoption of deep learning within mobile devices. The partnership gives Movidius access to Google's neural network technology roadmap, while the Search giant will source Movidius' processors and entire software development network.

"What Google has been able to achieve with neural networks is providing us with the building blocks for machine intelligence, laying the groundwork for the next decade of how technology will enhance the way people interact with the world," said Blaise Aguera y Arcas, head of Google's machine intelligence group. Arcas said that working with Movidius has allowed Google to expand its technology out of data centres and into the real world.

Google is using the MA2450, which is the most powerful iteration of Movidius'

Myriad 2 chip for this purpose. According to Remi El-Ouazzane, CEO, Movidius, the biggest challenge in embedding the technological advances Google has made in machine intelligence is in extreme power efficiency. This needs deep synthesis between the underlying hardware architecture and that is where the neural computer comes in.

In an interview with Digit, David Silver, Research Scientist on Google's Deep Mind, said that it's the early days for Artificial Intelligence and we are "decades away from human level AI". Silver heads the team that developed Deep Mind's AlphaGo algorithm, which recently beat Go champion Lee Seedol, in a best of five tournament. Dashwood says that machine learning and VPUs go hand-in-hand.

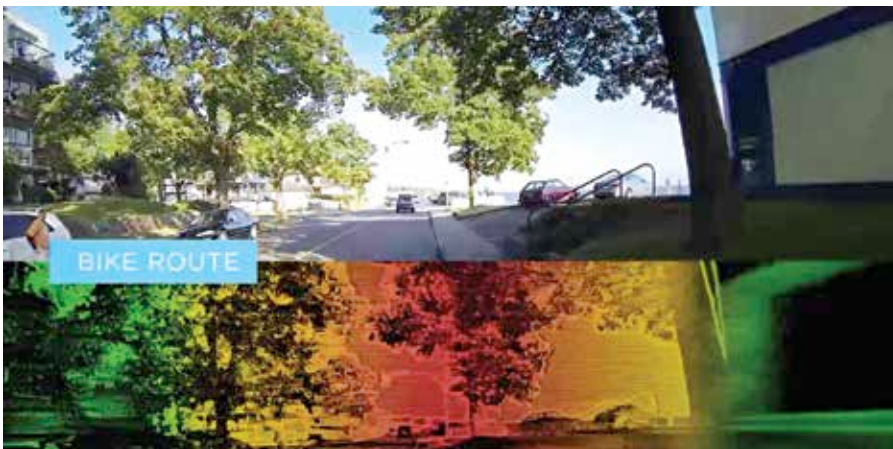
VPUs will, in future, make for an integral part of artificially intelligent robots, working as the eyes for the neural networks to work with.

The booming market

Adding the proverbial cherry on the cake, the Computer Vision and VPU market is at a nascent stage, but it's booming. Google and DJI are two of the best and biggest known names, but there are still many others exploring these avenues. Dashwood says that currently, Movidius, is the only viable solution in markets right now that presents low power, low thermal characteristics and high performance.

According to him, the company's chief competitors come from the GPU and CPU market, as in some cases, they may make for viable solutions for computer vision requirements. "In some instances, a CPU or GPU might make for a viable solution for high performance, OR, low power, OR, low thermal characteristics... but all three at the same time? We think we are the only viable solution right now," said he.

The Movidius MA2450 mentioned above is the only commercial solution for computer vision available on the market today. While VPUs won't offset chip-makers like Qualcomm, MediaTek and Intel, and they also won't compete against NVIDIA and AMD in the GPU segment either, they are however creating a whole new segment for themselves. 



Recreate a virtual replica of the real world with the aid of VPUs